

ESBJERG AP, Denmark

WMO: 060800

Lat: 55.528N

Lon: 8.563E

Elev: 30

StdP: 100.97

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-6.1	-10.9	1.5	-6.4	-8.8	1.8	-4.8	15.6	6.6	14.1	6.5	3.1	50	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.2	26.8	18.7	24.8	18.0	22.8	17.0	19.7	24.2	18.7	22.9	17.7	21.4	4.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	21.6	17.1	12.3	20.3	16.2	11.6	19.3	56.4	24.4	53.1	23.1	50.2	21.6	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.9	9.7	-10.6	29.4	3.5	2.0	-13.2	30.8	-15.3	32.0	-17.2	33.1	-19.8	34.6		
(5)			-11.0	21.7	3.4	1.1	-13.5	22.5	-15.5	23.2	-17.4	23.9	-19.9	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	1.6	1.8	3.7	7.7	11.3	14.1	16.6	16.7	13.9	10.0	5.9	3.1
	DBStd	6.31	3.83	3.69	3.14	3.33	3.29	2.75	2.71	2.52	2.48	2.99	3.40	3.99
	HDD10.0	1176	262	229	196	86	22	1	0	0	2	37	127	213
	HDD18.3	3481	520	463	454	321	219	132	71	64	134	258	373	472
	CDD10.0	772	0	0	0	16	63	123	203	208	118	37	3	0
	CDD18.3	37	0	0	0	0	1	4	16	14	2	0	0	0
	CDH23.3	307	0	0	0	1	12	42	139	107	7	0	0	0
	CDH26.7	49	0	0	0	0	1	3	24	20	1	0	0	0
Wind	WSAvg	4.7	5.2	5.5	5.4	4.7	4.6	4.6	4.1	4.0	4.4	4.6	4.7	5.0
Precipitation	PrecAvg	889	80	58	52	44	48	65	71	90	93	102	90	91
	PrecMax	1116	137	128	105	90	104	129	125	175	202	248	179	184
	PrecMin	585	3	16	8	5	3	1	16	20	24	32	24	20
	PrecStd	137	35	29	28	21	23	32	32	40	44	44	40	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	9.0	13.9	21.2	24.9	27.0	29.1	29.0	24.2	18.0	13.9	10.9
		MCWB	8.3	7.9	9.4	13.4	17.4	18.2	19.3	18.1	18.3	15.4	12.7	9.8
	2%	DB	8.2	8.0	10.9	18.5	22.2	24.2	26.9	26.8	21.5	16.1	12.6	9.9
		MCWB	7.4	7.3	7.8	12.0	15.1	17.1	19.1	18.5	16.9	14.3	11.5	9.4
	5%	DB	7.3	7.1	9.0	16.0	20.0	22.0	24.9	24.0	19.2	15.1	11.4	9.0
		MCWB	6.6	6.5	7.2	11.0	14.0	16.4	18.1	18.0	15.9	13.5	10.2	8.3
	10%	DB	6.8	6.2	7.9	13.2	17.8	19.8	22.2	21.9	17.9	14.1	10.3	8.1
		MCWB	6.1	5.7	6.5	9.6	12.9	15.3	17.2	17.3	15.2	12.7	9.2	7.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.6	8.1	10.0	15.0	18.5	20.0	21.2	21.1	19.5	15.7	13.0	10.1
		MCDB	9.1	8.4	11.9	19.5	23.5	24.4	26.6	25.3	22.7	17.2	13.7	10.4
	2%	WB	7.5	7.2	8.3	12.7	16.1	18.3	19.9	19.5	17.6	14.9	11.7	9.2
		MCDB	7.9	7.6	10.0	17.0	20.5	22.8	25.0	23.3	20.2	16.0	12.6	9.5
	5%	WB	6.6	6.5	7.3	11.4	14.8	16.9	18.7	18.5	16.6	13.9	10.6	8.2
		MCDB	7.1	6.9	8.3	15.0	18.7	20.7	23.1	22.4	18.7	14.9	11.4	8.6
	10%	WB	5.7	5.8	6.6	10.1	13.4	15.6	17.7	17.6	15.7	13.1	9.5	7.4
		MCDB	6.2	6.2	7.3	12.8	17.0	18.9	21.4	21.2	17.4	14.2	10.2	7.9
Mean Daily Temperature Range	5% DB	MDBR	4.1	4.5	6.0	8.3	8.4	7.8	8.2	8.2	7.0	5.7	4.6	4.3
		MCDBR	3.5	4.3	7.9	12.2	12.6	12.6	12.8	12.0	9.2	6.1	4.0	3.6
		MCWBR	3.7	3.8	5.6	7.1	7.1	7.2	6.4	5.7	5.7	4.8	4.0	3.7
	5% WB	MCDBR	3.4	3.9	6.4	10.6	10.8	10.8	11.1	9.9	7.8	5.5	3.9	3.5
		MCWBR	3.7	3.6	5.0	6.8	6.8	6.9	6.2	5.7	5.3	4.7	4.1	3.7
Clear-Sky Solar Irradiance	taub	0.323	0.340	0.362	0.395	0.386	0.383	0.396	0.398	0.378	0.357	0.333	0.310	
	taud	2.247	2.298	2.299	2.257	2.343	2.373	2.351	2.355	2.396	2.381	2.302	2.245	
	Ebn at Noon	577	722	799	823	856	862	840	813	778	696	563	501	
	Edh at Noon	58	78	98	118	115	113	113	106	90	72	54	46	
All-Sky Solar Radiation	RadAvg	0.49	1.11	2.43	4.05	5.29	5.59	5.21	4.17	2.89	1.49	0.61	0.36	
	RadStd	0.08	0.15	0.22	0.48	0.52	0.43	0.55	0.30	0.30	0.18	0.09	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	-112	-160	N/A	N/A

Nomenclature: See separate page